Program : **Diploma in Fire Technology and Safety**	
Course Code: **6467**	Course Title: **Computer-Aided Risk Analysis Lab**
Semester: **6**	Credits: **2.5**
Course Category: **Program Core**	
Periods per week: **4 (L:0, T:1, P:3)**	Periods per semester: **45**

Course Objectives:

- To familiarize students with risk analysis techniques using computer-aided tools.
- To apply risk assessment models for fire safety and industrial hazards.
- To analyze and interpret risk factors using simulation software.
- To develop competency in hazard identification, risk evaluation, and mitigation strategies.

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Understand the concepts and methodologies of risk analysis.	9	Understanding
CO2	Use computer software to conduct hazard identification and risk assessment.	11	Applying
CO3	Interpret simulation results for industrial safety applications.	11	Analyzing
CO4	Develop risk mitigation strategies based on computer-aided analysis.	10	Evaluating
	Lab Exam	4	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2	1					
CO2		2	1				
CO3		1				1	2
CO4						2	

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Understand the concepts and methodologies of risk analysis.**		
M1.01	Overview of risk assessment methods	3	Understanding
M1.02	Introduction to risk analysis software tools	6	Applying
CO2	**Use computer software to conduct hazard identification and risk assessment.**		
M2.01	Study of hazard identification techniques	5	Applying
M2.02	Application of HAZOP, FMEA, and FTA using software	6	Applying
	Lab Exam I	2	
CO3	**Interpret simulation results for industrial safety applications.**		
M3.01	Conducting quantitative risk assessment (QRA)	5	Applying
M3.02	Simulation and interpretation of risk assessment data	6	Analyzing
CO4	**Develop risk mitigation strategies based on computer-aided analysis.**		
M4.01	Development of risk control and mitigation plans	6	Applying
M4.02	Case study analysis of industrial safety incidents	4	Evaluating
	Lab Exam – II	2	